Cálculo de coeficientes de la Serie de Fourier

Coeficientes an:

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} x\,dx$$

$$= \frac{1}{\pi} * \frac{x^2}{2}\Big|_{-\pi}^{\pi} = \frac{1}{\pi}\left(\frac{\pi^2}{2} - \frac{-\pi^2}{2}\right) = \frac{1}{\pi}\left(\frac{\pi^2}{2} - \frac{\pi^2}{2}\right)$$

$$= 0 \quad para\ n \neq 0$$

Coeficiente a0:

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} x\cos\left(\frac{n\pi x}{\pi}\right)dx = \frac{1}{\pi}\int_{-\pi}^{\pi} x\cos(nx)\,dx$$

$$= \frac{1}{\pi}\left(\left(\frac{1}{n}x\ \sin nx\right)\Big|_{-\pi}^{\pi} - \frac{1}{n}\int_{-\pi}^{\pi}\sin(nx)\,dx\right)$$

$$= \frac{1}{\pi}\left(\left(\frac{1}{n}\pi\sin n\pi\right) - \left(\frac{1}{n}(-\pi)\sin n(\pi)\right) + \frac{1}{n^2}\cos(nx)\Big|_{-\pi}^{\pi}\right)$$

$$= \frac{1}{\pi}\left(\left(\frac{1}{n}\pi\sin n\pi\right) - \left(\frac{1}{n}(-\pi)\sin n(-\pi)\right) + \frac{1}{n^2}\cos(n\pi) - \frac{1}{n^2}\cos\big(n(-\pi)\big)\right)$$

$$= 0$$

Coeficiente bn

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} x\sin(nx)\,dx$$

$$= \frac{1}{\pi}\left(\left(-\frac{1}{n}x\ \cos nx\right)\Big|_{-\pi}^{\pi} + \frac{1}{n}\int_{-\pi}^{\pi}\cos(nx)\,dx\right)$$

$$\frac{1}{\pi}\left(\left(\frac{1}{n}\pi\cos n\pi\right) + \left(\frac{1}{n}(-\pi)\cos n(-\pi)\right) + \frac{1}{n^2}\sin(nx)\Big|_{-\pi}^{\pi}\right)$$

$$\frac{1}{\pi}\left(\left(-\frac{2}{n}\pi\cos n\pi\right) + \frac{1}{n^2}\sin(nx) - \frac{1}{n^2}\sin(-nx)\right)$$

$$\frac{1}{\pi}\left(\left(-\frac{2}{n}\pi\cos n\pi\right)\right)$$

$$\cos(n\pi) = (-1)^n$$

$$\left(-\frac{2}{n}(-1)^n\right) = \frac{2}{n}(-1)^{n+1}$$

Entonces

$$x = \sum_{n=1}^{\infty} \frac{2}{n}(-1)^{n+1} \sin{(nx)}$$

Gráfica en Matlab con 50 elementos: